Sprint 02 Assignment: Regression Modelling

Basil Oforbuike Emeokoro
ADA Global Academy MicroMasters in Machine Learning and Research Methods

Research Question 1: What area−level economic and housing characteristics are associated with house price?
Research Question 2: What patient and hospital−stay characteristics are associated with diabetes readmission?

Dataset	Rows	Columns	Target	Task
USA Housing	5000	7	Price	Multiple linear regression
Diabetes readmission	101766	16	readmitted_binary	Binary logistic regression

Dataset Documentation and Preparation

Table 1. Dataset overview

Dataset	Rows	Columns	Target	Task
USA Housing	5000	7	Price	Multiple linear regression
Diabetes readmission	101766	16	readmitted_binary	Binary logistic regression

Table 2. Missing and duplicate checks

Dataset	Missing_Cells	Duplicate_Rows
USA Housing	0	0
Diabetes readmission	0	35

Decisions:

USA Housing uses Price as the continuous outcome. Address is excluded because it is a text identifier rather than a stable numeric predictor for the assignment model.

Diabetes readmitted_binary is converted to 0/1. Race, gender, and age are treated as factors. Medical specialty and *_id administrative codes are documented but excluded from the core model to preserve interpretability and stability.

Variable Descriptions

Table 3. Variable descriptions

Dataset	Variable	Description
USA Housing	Avg. Area Income	Average income in the area.
USA Housing	Avg. Area House Age	Average house age in the area.
USA Housing	Avg. Area Number of Rooms	Average number of rooms in houses in the area.
USA Housing	Avg. Area Number of Bedrooms	Average number of bedrooms in houses in the area.
USA Housing	Area Population	Area population.
USA Housing	Price	House price; continuous target variable.
USA Housing	Address	Text address; excluded because it is non−predictive text for this assignment model.
Diabetes readmission	time_in_hospital	
Diabetes readmission	num_lab_procedures	
Diabetes readmission	num_procedures	
Diabetes readmission	num_medications	
Diabetes readmission	number_outpatient	
Diabetes readmission	number_emergency	
Diabetes readmission	number_inpatient	
Diabetes readmission	number_diagnoses	
Diabetes readmission	race	
Diabetes readmission	gender	
Diabetes readmission	age	
Diabetes readmission	admission_type_id	
Diabetes readmission	discharge_disposition_id	
Diabetes readmission	admission_source_id	
Diabetes readmission	medical_specialty	
Diabetes readmission	readmitted_binary	

Length of hospital stay.
Number of laboratory procedures.
Number of non−laboratory procedures.
Number of medications.
Number of outpatient visits before encounter.
Number of emergency visits before encounter.
Number of inpatient visits before encounter.
Number of diagnoses.
Patient race category.
Patient gender.
Patient age band.
Admission type code; administrative coded variable.
Discharge disposition code; administrative coded variable.
Admission source code; administrative coded variable.
Medical specialty associated with encounter.
Binary readmission target.

Exploratory Data Analysis Tables

Table 4. Housing descriptive statistics

Dataset	Variable	N	Mean	SD	Min	Median	Max
USA Housing	Price	5000	1.232e+06	3.531e+05	15938.658	1.233e+06	2.469e+06
USA Housing	Avg_Area_Income	5000	6.858e+04	1.066e+04	17796.631	6.880e+04	1.077e+05
USA Housing	Avg_Area_House_Age	5000	5.977e+00	9.915e−01	2.644	5.970e+00	9.519e+00
USA Housing	Avg_Area_Number_of_Rooms	5000	6.988e+00	1.006e+00	3.236	7.003e+00	1.076e+01
USA Housing	Avg_Area_Number_of_Bedrooms	5000	3.981e+00	1.234e+00	2.000	4.050e+00	6.500e+00
USA Housing	Area_Population	5000	3.616e+04	9.926e+03	172.611	3.620e+04	6.962e+04

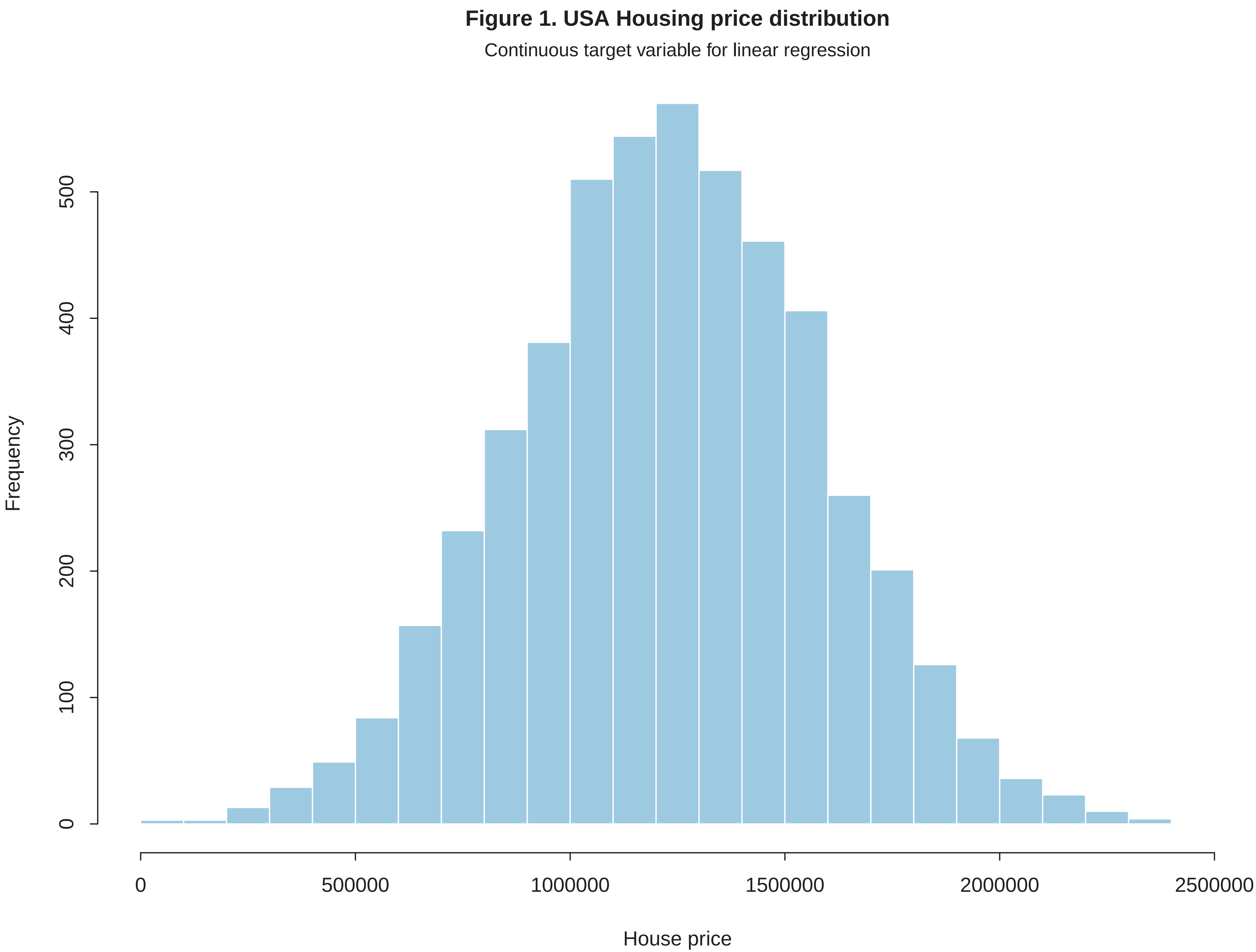
Table 5. Diabetes descriptive statistics

Dataset	Variable	N	Mean	SD	Min	Median	Max
Diabetes readmission	time_in_hospital	101766	4.3960	2.9851	1	4	14
Diabetes readmission	num_lab_procedures	101766	43.0956	19.6744	1	44	132
Diabetes readmission	num_procedures	101766	1.3397	1.7058	0	1	6
Diabetes readmission	num_medications	101766	16.0218	8.1276	1	15	81
Diabetes readmission	number_outpatient	101766	0.3694	1.2673	0	0	42
Diabetes readmission	number_emergency	101766	0.1978	0.9305	0	0	76
Diabetes readmission	number_inpatient	101766	0.6356	1.2629	0	0	21
Diabetes readmission	number_diagnoses	101766	7.4226	1.9336	1	8	16

Table 6. Readmission distribution

Outcome	N	Percent
Not readmitted	54864	53.91
Readmitted	46902	46.09

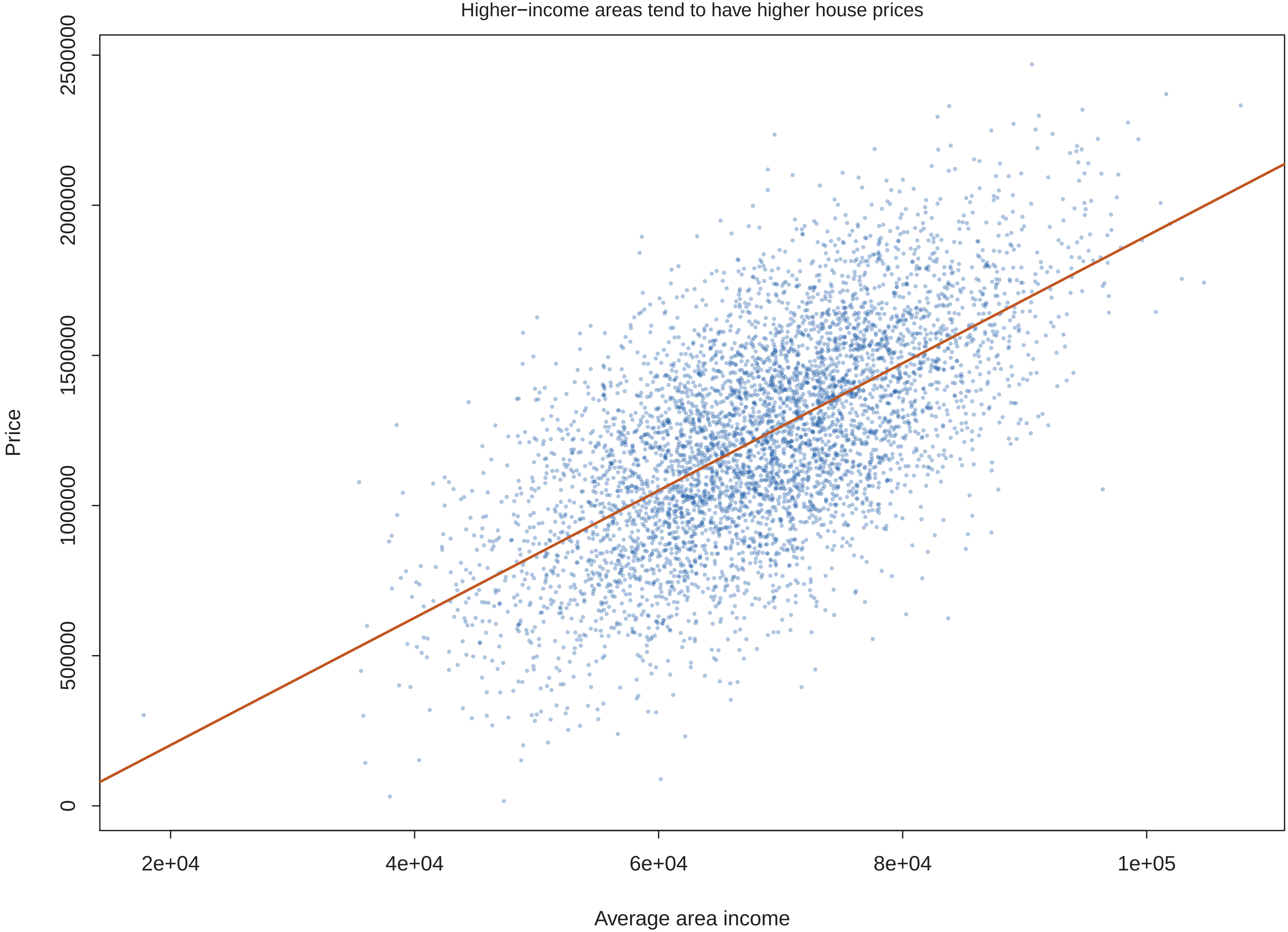
Housing price is most visibly related to average area income, house age, and rooms. Diabetes readmission appears higher among patients with more prior inpatient visits and somewhat longer hospital stays.



Caption: Price is approximately bell−shaped with high−value upper−tail observations.

Figure 2. Price vs average area income

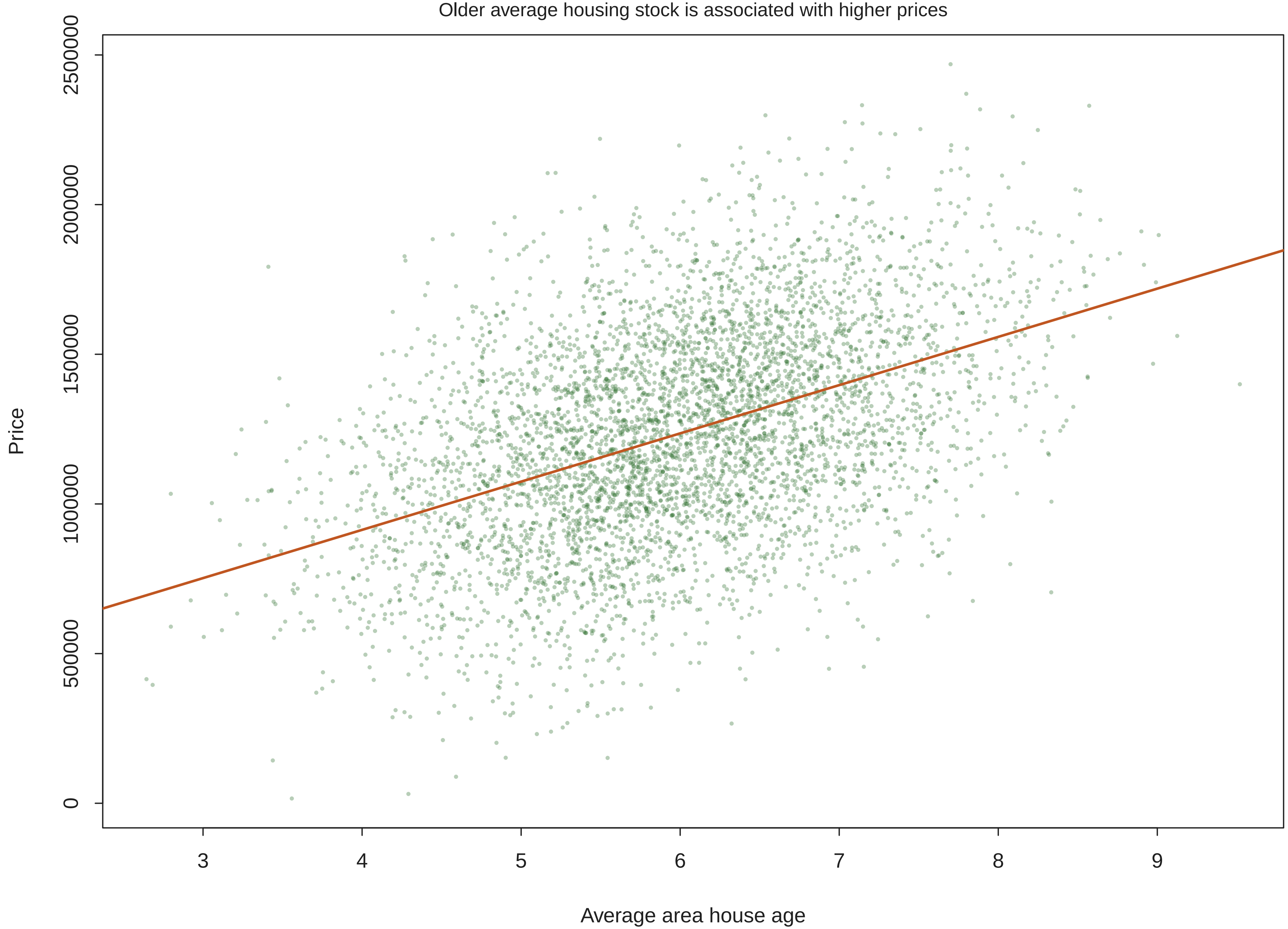
Higher−income areas tend to have higher house prices



Caption: The fitted line shows a strong positive bivariate relationship.

Figure 3. Price vs house age

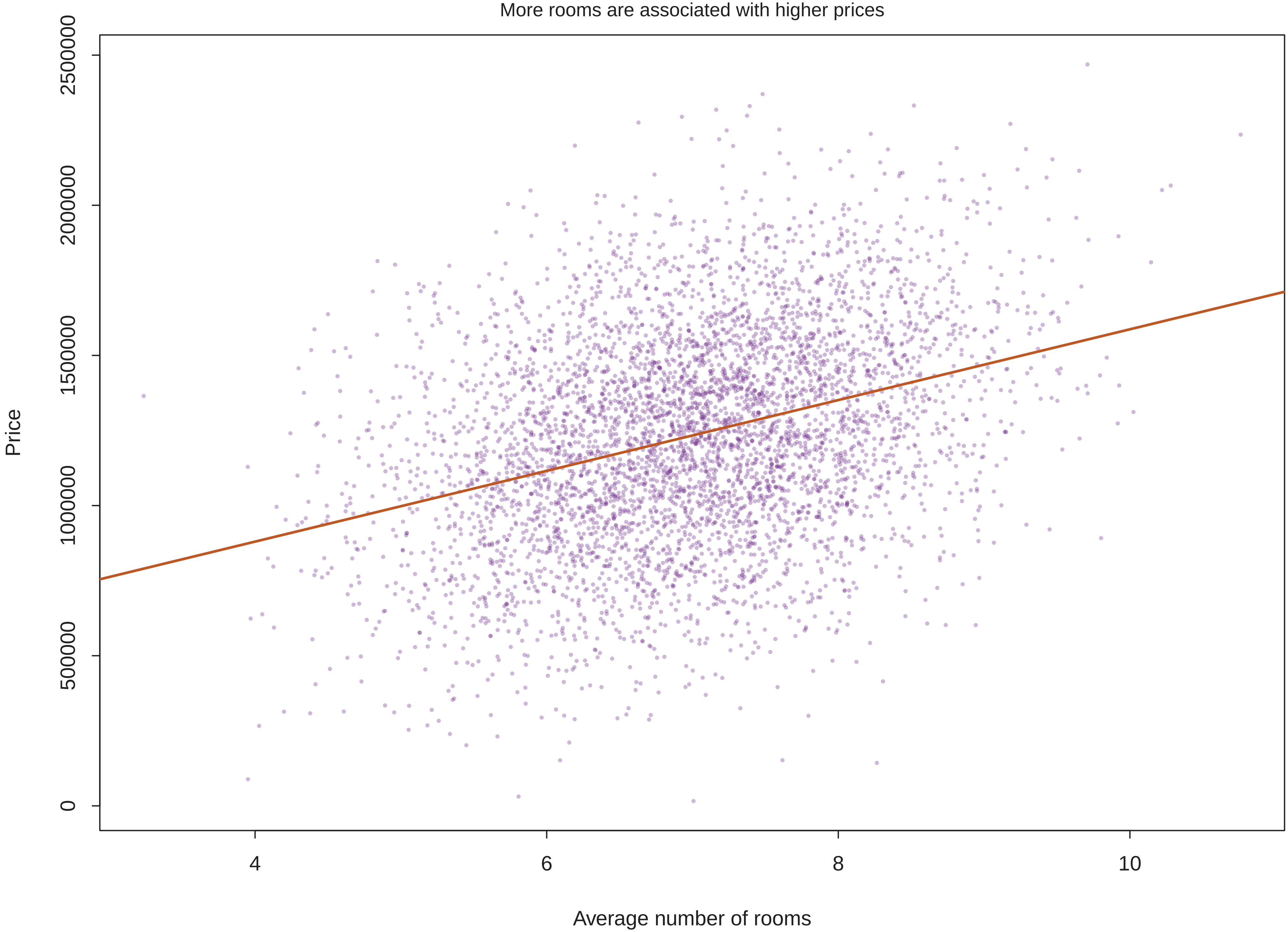
Older average housing stock is associated with higher prices



Caption: Relationship is positive in the assignment dataset.

Figure 4. Price vs average rooms

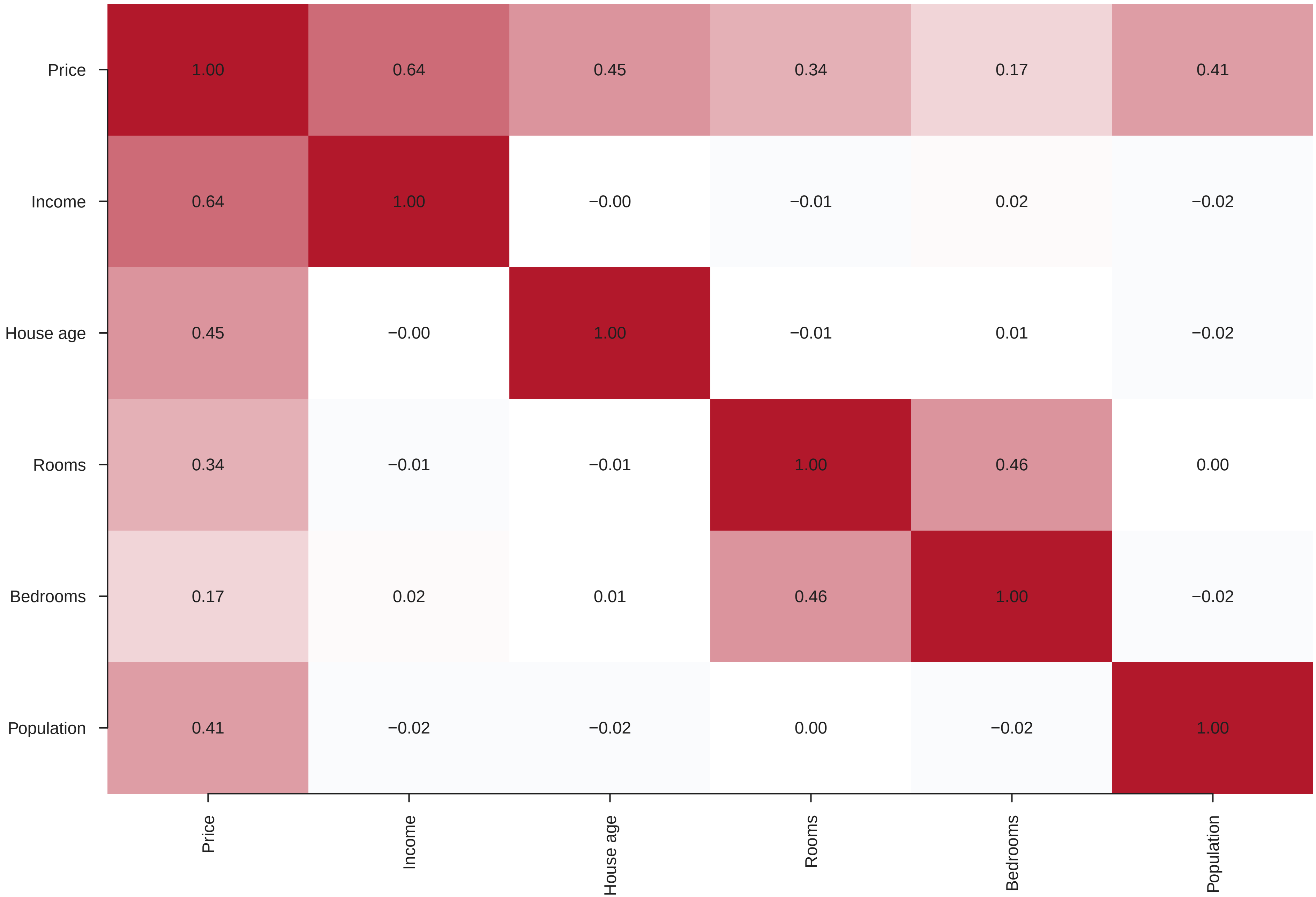
More rooms are associated with higher prices



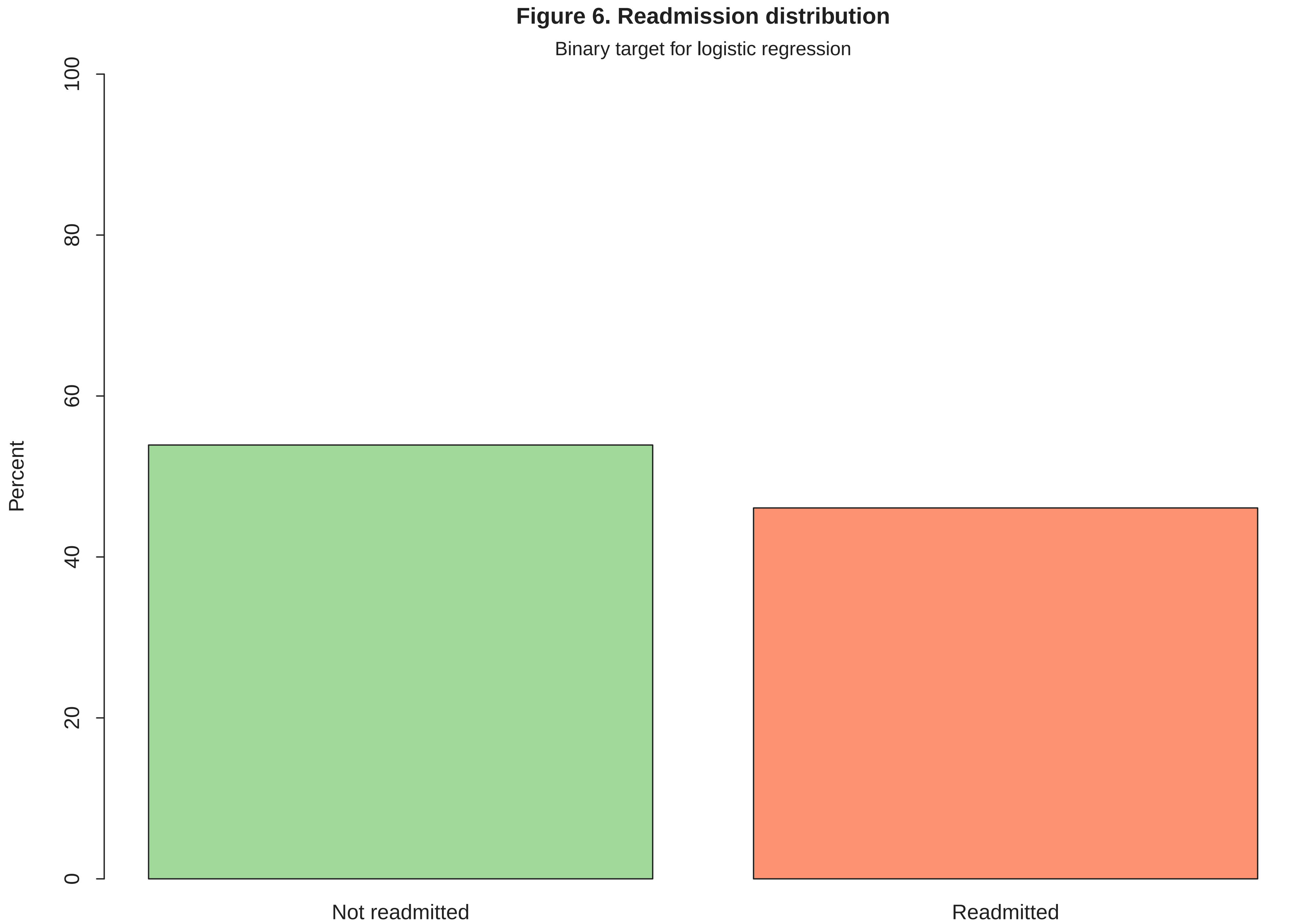
Caption: Room count has a positive association with price.

Figure 5. Housing correlation heatmap

Red indicates positive association; blue indicates negative



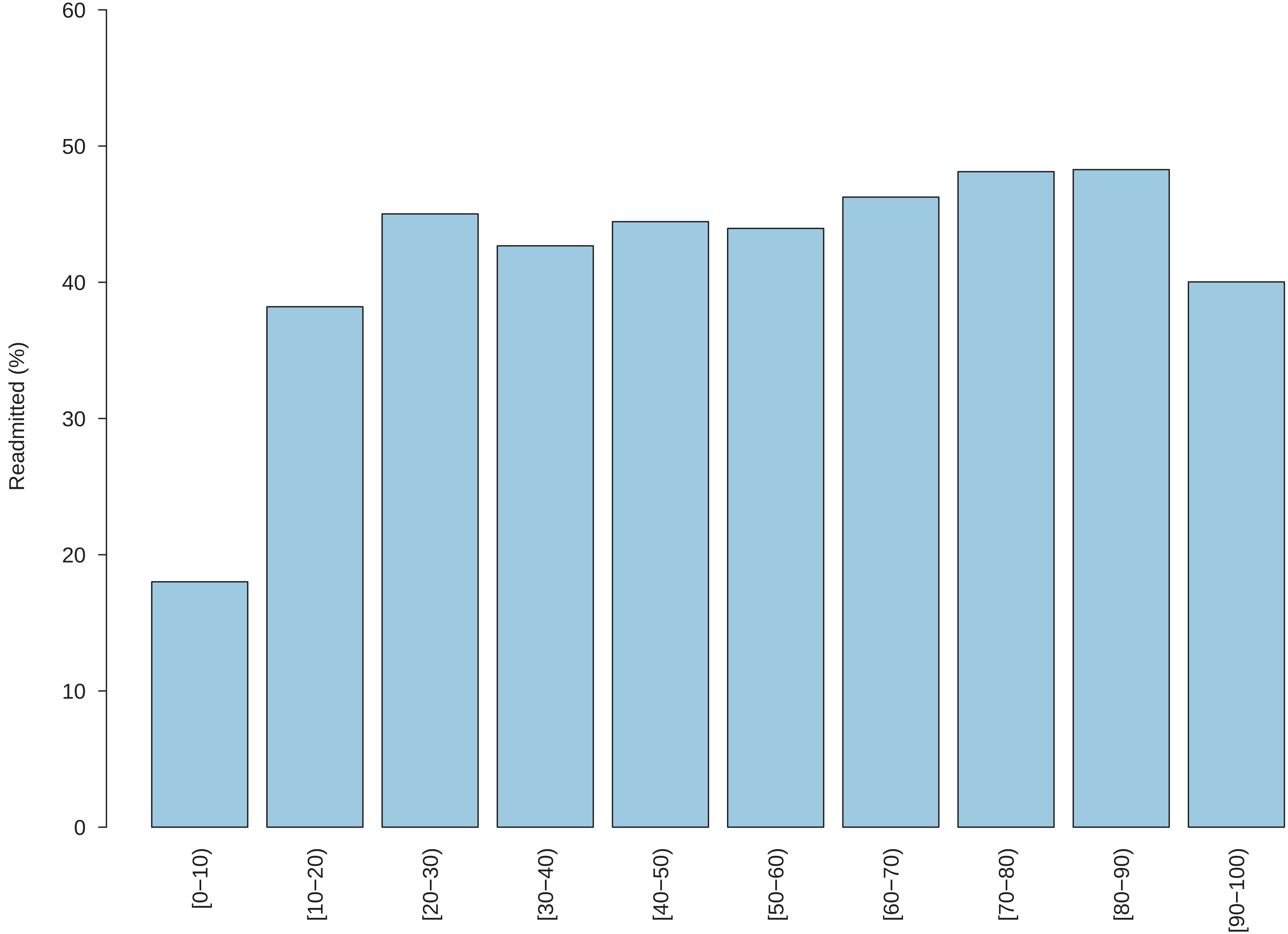
Caption: Income and room–related predictors show the clearest links with price.



Caption: Class balance affects how accuracy should be interpreted.

Figure 7. Readmission by age group

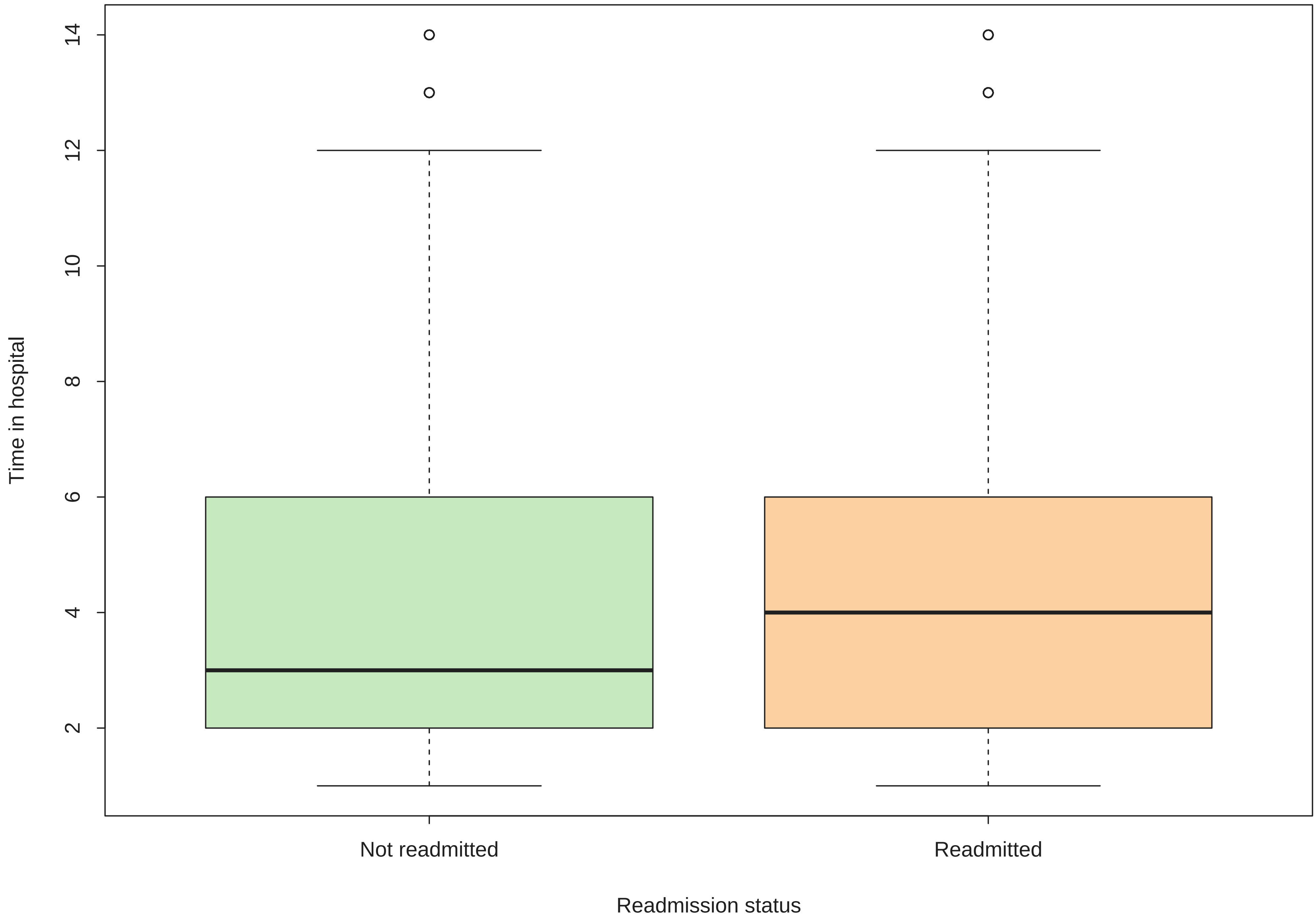
Older age bands generally have higher readmission percentages



Caption: Percent readmitted is calculated within each age group

Figure 8. Readmission by time in hospital

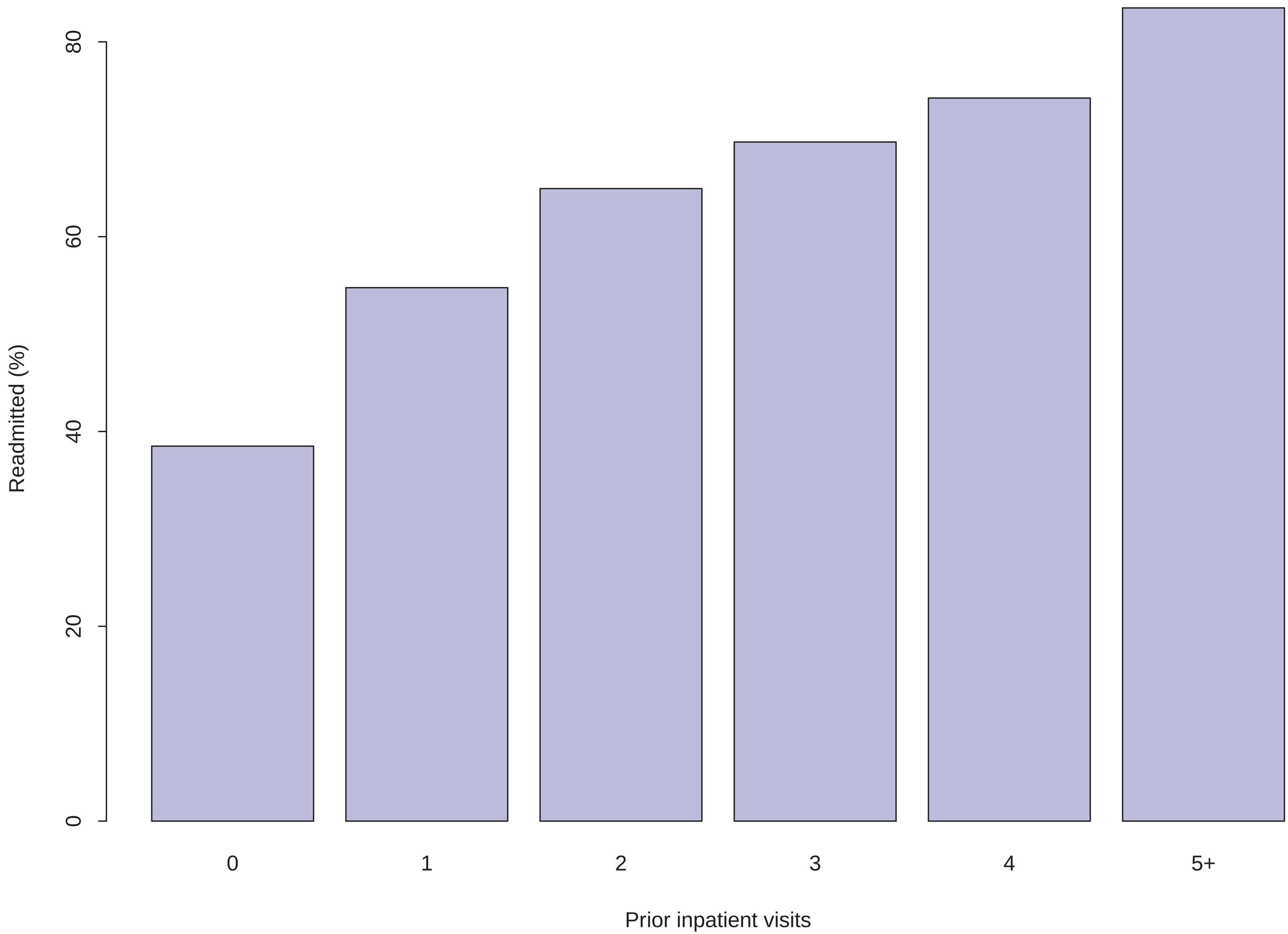
Readmitted patients have slightly longer stays



Caption: Box plots compare hospital stay length by outcome.

Figure 9. Readmission by inpatient visits

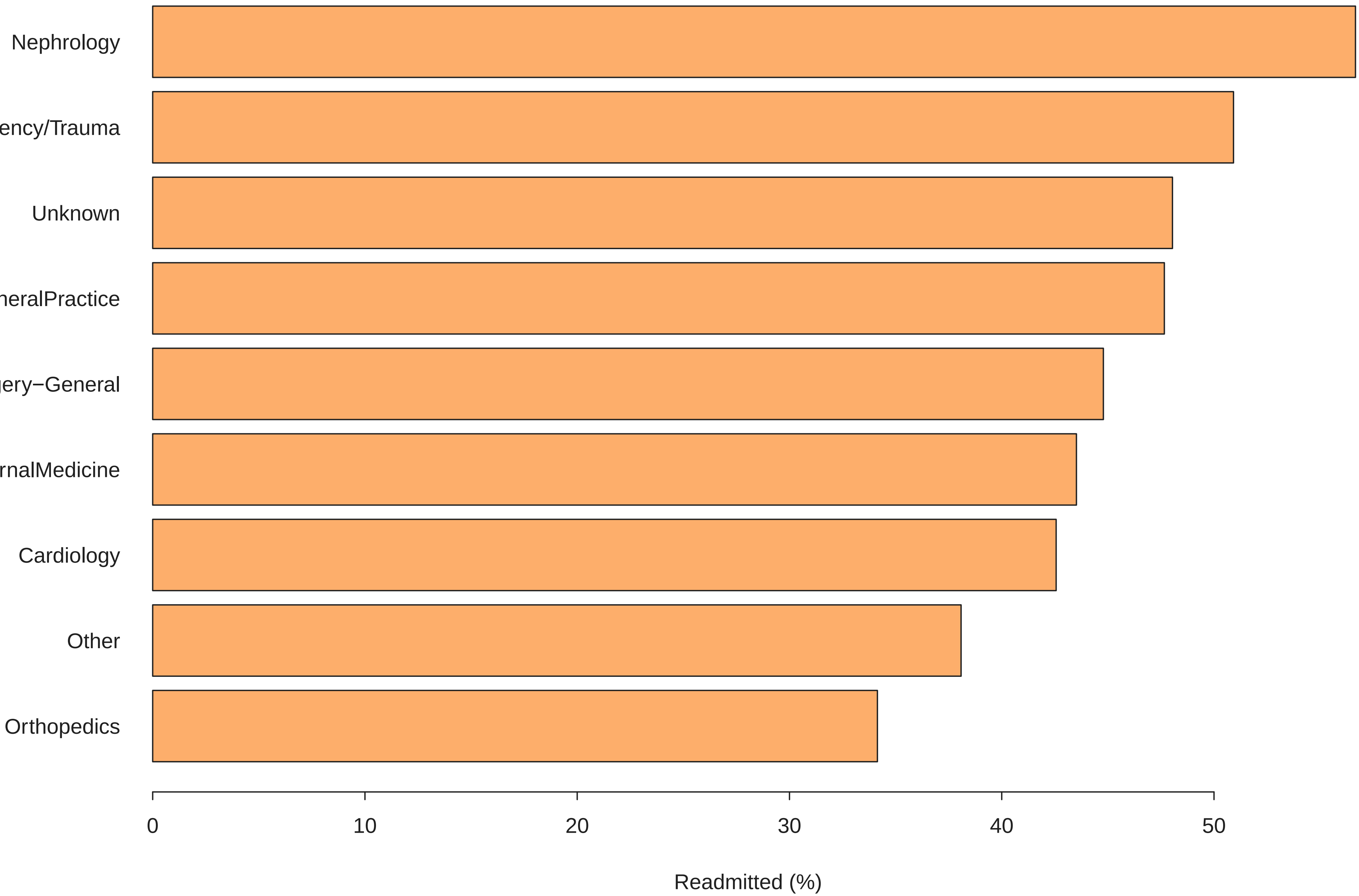
Prior inpatient use is strongly associated with readmission



Caption: Counts of 5 or more prior inpatient visits are grouped as 5+.

Figure 10. Readmission by medical specialty

Healthcare specialty captures patient and care−context differences



Caption: Smaller specialties are grouped as Other.

Linear Regression Results

Model formula:
Price ~ Avg. Area Income + Avg. Area House Age + Avg. Area Number of Rooms + Avg. Area Number of Bedrooms + Area Population

Table 7. Linear regression coefficients

Term	Estimate	Std_Error	t_value	p_value	CI_2.5	CI_97.5
(Intercept)	−2.637e+06	1.716e+04	−153.708	0.0000	−2.671e+06	−2.604e+06
Area_Population	1.520e+01	1.442e−01	105.393	0.0000	1.492e+01	1.548e+01
Avg_Area_House_Age	1.656e+05	1.443e+03	114.754	0.0000	1.628e+05	1.685e+05
Avg_Area_Income	2.158e+01	1.343e−01	160.656	0.0000	2.131e+01	2.184e+01
Avg_Area_Number_of_Bedrooms	1.651e+03	1.309e+03	1.262	0.2071	−9.144e+02	4.217e+03
Avg_Area_Number_of_Rooms	1.207e+05	1.605e+03	75.170	0.0000	1.175e+05	1.238e+05

Table 8. Fit metrics

Metric	Value
R−squared	9.180e−01
Adjusted R−squared	9.179e−01
Residual standard error	1.012e+05
Model N	5.000e+03

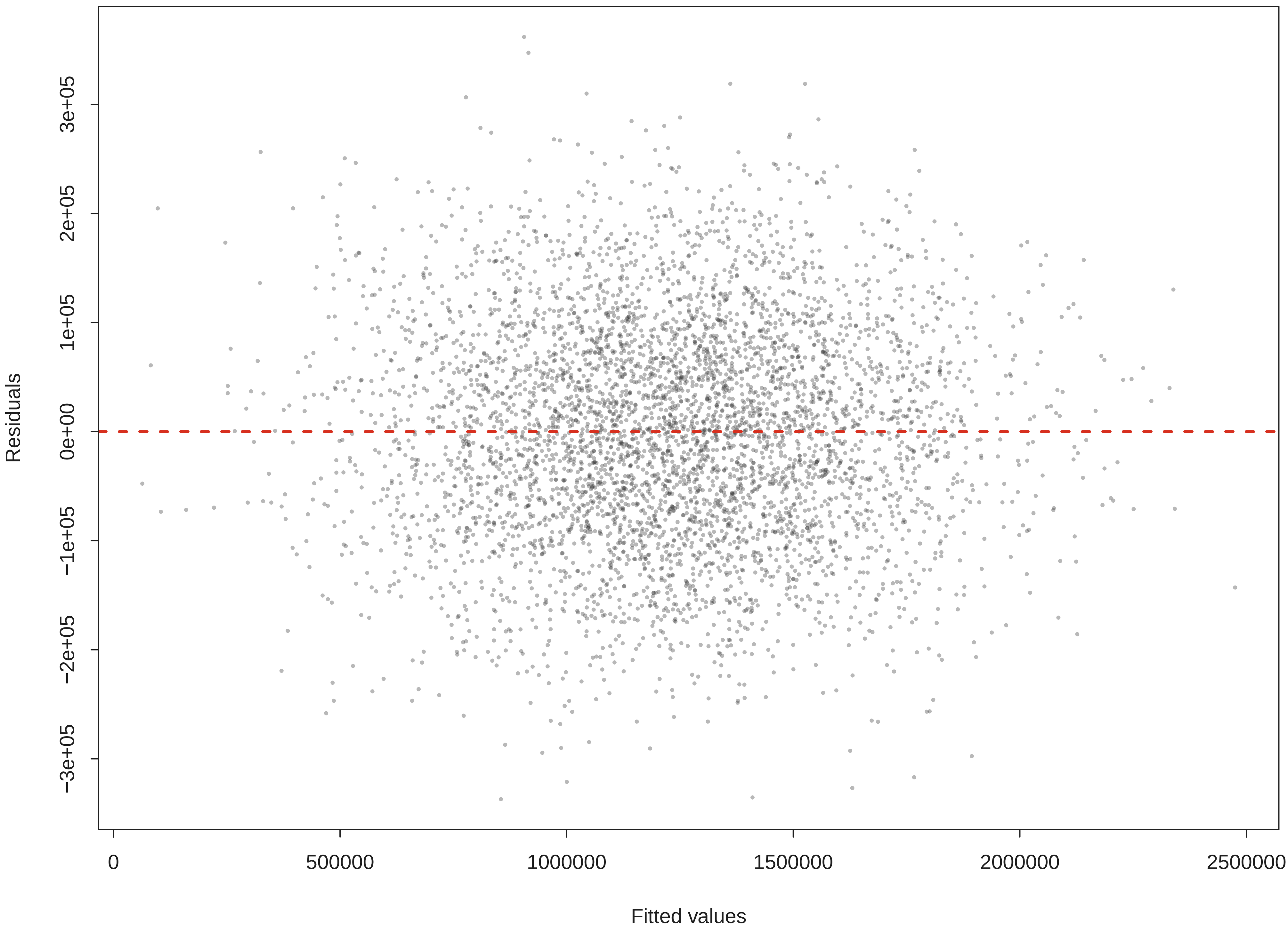
Table 9. Prediction interval for median−profile house

Avg_Area_Income	Avg_Area_House_Age	Avg_Area_Number_of_Rooms	Avg_Area_Number_of_Bedrooms	Area_Population	fit	lwr	upr
68804	5.97	7.003	4.05	36199	1238202	1039877	1436527

The multiple linear regression explains 91.8% of variation in Price (adjusted R−squared 0.918). Holding the other model variables constant, positive coefficients indicate higher expected price. The largest coefficient by scale is Avg_Area_House_Age, but variable units differ, so practical importance should be judged with context. Statistically significant predictors at $p < 0.05$ are: Area_Population, Avg_Area_House_Age, Avg_Area_Income, Avg_Area_Number_of_Rooms.

Figure 11. Residuals vs fitted values

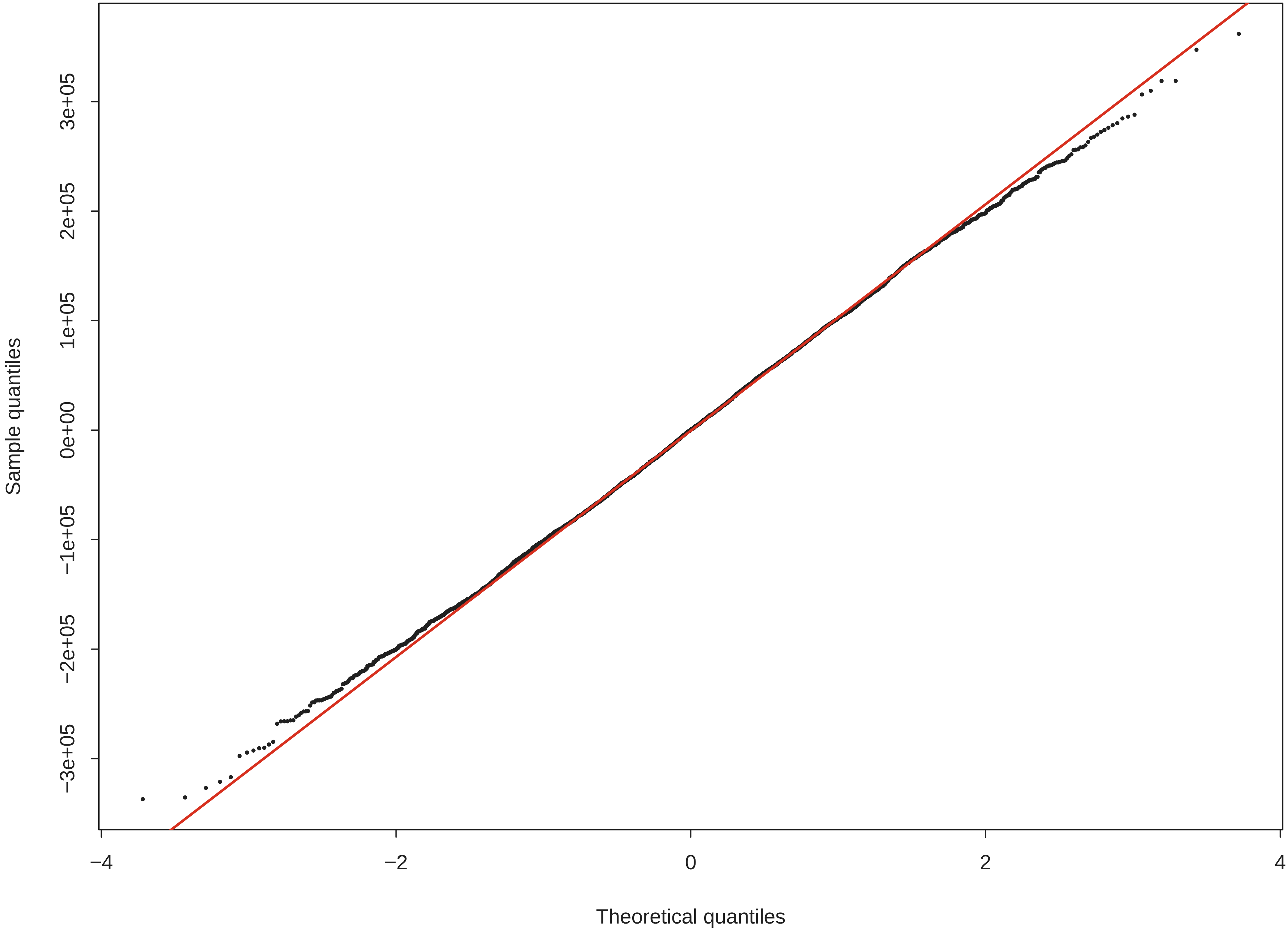
Linear regression diagnostic



Caption: Residual spread is used to assess linearity and constant variance.

Figure 12. Normal Q−Q plot

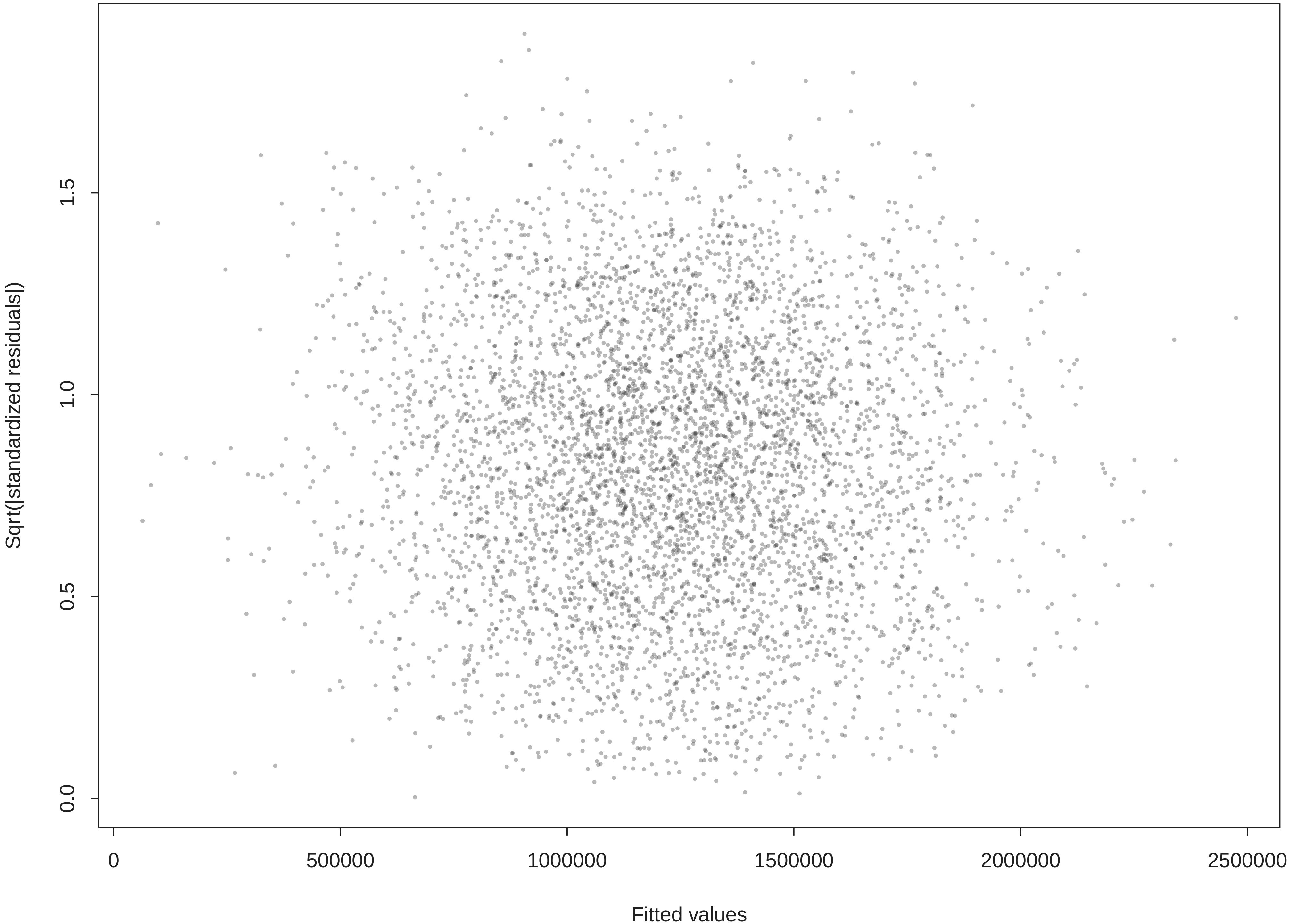
Linear regression diagnostic



Caption: Points close to the line indicate approximately normal residuals.

Figure 13. Scale−location plot

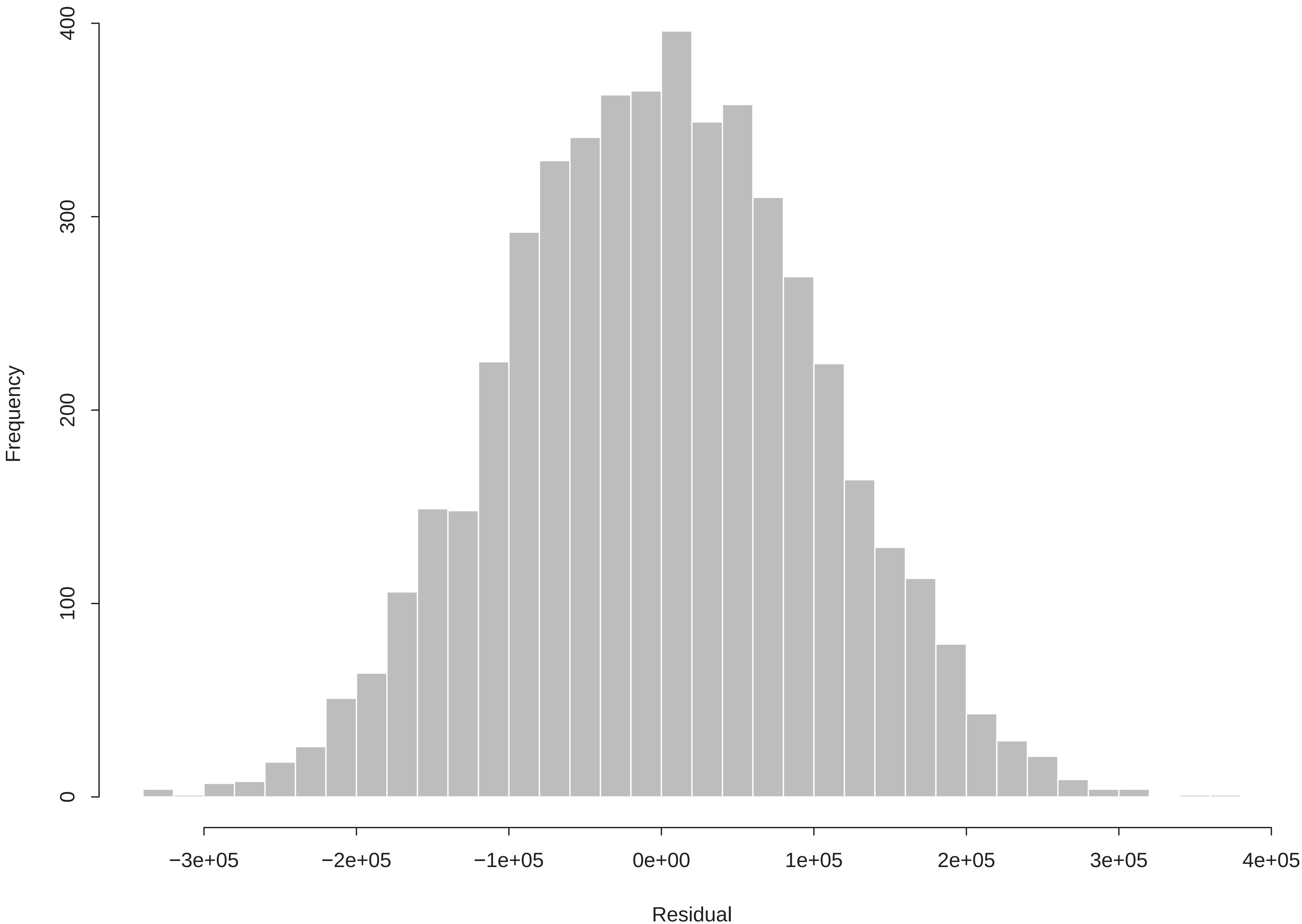
Linear regression diagnostic



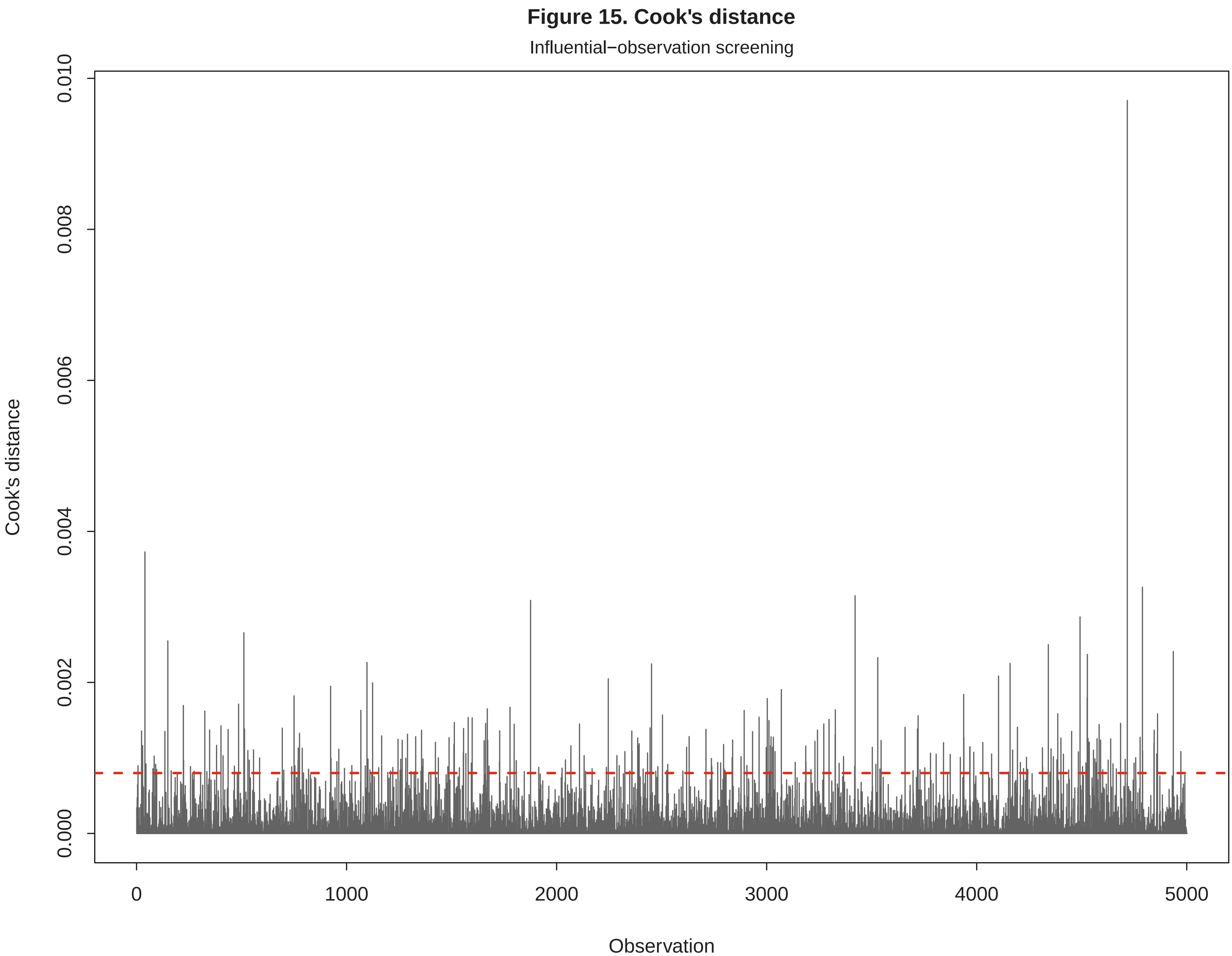
Caption: A flat spread suggests more stable residual variance.

Figure 14. Residual histogram

Distribution of linear regression residuals



Caption: Histogram supports the Q−Q plot assessment.



Caption: Dashed line shows the common 4/n screening threshold.

Linear Regression Diagnostics

Table 10. Manual VIF values

Variable	VIF
Avg_Area_Income	1.001
Avg_Area_House_Age	1.001
Avg_Area_Number_of_Rooms	1.274
Avg_Area_Number_of_Bedrooms	1.274
Area_Population	1.001

Residual plots are used to assess linearity, variance stability, residual normality, and influential observations. VIF values summarize multicollinearity among the numeric housing predictors. None of these checks proves causality; they indicate whether the linear model is a reasonable descriptive approximation.

Logistic Regression Results

```
Model formula:
readmitted_binary ~ time_in_hospital + num_lab_procedures + num_procedures +
                    num_medications + number_outpatient + number_emergency +
                    number_inpatient + number_diagnoses + race + gender + age
```

Table 11. Logistic coefficients and odds ratios

Term	Estimate	Std_Error	z_value	p_value	Odds_Ratio	OR_CI_2.5	OR_CI_97.5
(Intercept)	−1.939119	2.545e−01	−7.6187	2.562e−14	0.1438307	8.734e−02	2.369e−01
age[10−20)	0.836930	2.715e−01	3.0826	2.052e−03	2.3092673	1.356e+00	3.932e+00
age[20−30)	0.770775	2.615e−01	2.9481	3.198e−03	2.1614406	1.295e+00	3.608e+00
age[30−40)	0.695043	2.572e−01	2.7028	6.877e−03	2.0037946	1.210e+00	3.317e+00
age[40−50)	0.763498	2.552e−01	2.9920	2.771e−03	2.1457699	1.301e+00	3.538e+00
age[50−60)	0.778761	2.547e−01	3.0576	2.231e−03	2.1787703	1.323e+00	3.589e+00
age[60−70)	0.851610	2.546e−01	3.3450	8.228e−04	2.3434159	1.423e+00	3.860e+00
age[70−80)	0.910008	2.545e−01	3.5754	3.497e−04	2.4843414		
age[80−90)	0.860840	2.548e−01	3.3786	7.285e−04	2.3651474		
age[90−100)	0.500335	2.586e−01	1.9350	5.299e−02	1.6492729		
genderMale	−0.049355	1.582e−02	−3.1189	1.815e−03	0.9518430		
genderUnknown/Invalid	−8.648307	4.128e+01	−0.2095	8.341e−01	0.0001754		
num_lab_procedures	0.001227	4.243e−04	2.8921	3.827e−03	1.0012278		
num_medications	0.003661	1.200e−03	3.0505	2.285e−03	1.0036674		
num_procedures	−0.054966	5.068e−03	−10.8462	2.079e−27	0.9465170		
number_diagnoses	0.075931	4.455e−03	17.0450	3.808e−65	1.0788887		
number_emergency	0.234080	1.481e−02	15.8039	2.925e−56	1.2637450		
number_inpatient	0.363875	8.200e−03	44.3745	0.000e+00	1.4388944		

```
 1.509e+00 4.091e+00
 1.435e+00 3.897e+00
 9.936e−01 2.738e+00
 9.228e−01 9.818e−01
 1.279e−39 2.407e+31
 1.000e+00 1.002e+00
 1.001e+00 1.006e+00
 9.372e−01 9.560e−01
 1.070e+00 1.088e+00
 1.228e+00 1.301e+00
 1.416e+00 1.462e+00
```

```
Top odds−ratio effects by absolute log odds ratio:
 Term                  Estimate Std_Error z_value p_value   Odds_Ratio
 genderUnknown/Invalid −8.6483  41.2797   −0.2095 0.8340538 0.0001754
 age[70−80)             0.9100   0.2545    3.5754 0.0003497 2.4843414
 age[80−90)             0.8608   0.2548    3.3786 0.0007285 2.3651474
 age[60−70)             0.8516   0.2546    3.3450 0.0008228 2.3434159
 age[10−20)             0.8369   0.2715    3.0826 0.0020517 2.3092673
 OR_CI_2.5 OR_CI_97.5
 1.279e−39 2.407e+31
 1.509e+00 4.091e+00
 1.435e+00 3.897e+00
 1.423e+00 3.860e+00
 1.356e+00 3.932e+00
```

The logistic model estimates the probability of readmission. Odds ratios above 1 indicate higher odds of readmission, while odds ratios below 1 indicate lower odds. At threshold 0.5, test accuracy is 61.8%, sensitivity is 38.9%, specificity is 81.4%, and manual AUC is 0.651. Statistically significant predictors at $p < 0.05$ include: age[10−20), age[20−30), age[30−40), age[40−50), age[50−60), age[60−70), age[70−80), age[80−90), genderMale, num_lab_procedures, num_medications, num_procedures, number_diagnoses, number_emergency, number_inpatient, number_outpatient, raceCaucasian, raceOther, raceUnknown, time_in_hospital.

Logistic Evaluation

Table 12. Evaluation metrics

Metric	Value
Train N	7.124e+04
Test N	3.053e+04
Threshold	5.000e−01
Accuracy	6.179e−01
Sensitivity	3.890e−01
Specificity	8.137e−01
AUC	6.513e−01
Readmitted test prevalence	4.612e−01

Table 13. Confusion matrix

Predicted	Actual	N
Not readmitted	Not readmitted	13387
Not readmitted	Readmitted	8602
Readmitted	Not readmitted	3064
Readmitted	Readmitted	5477

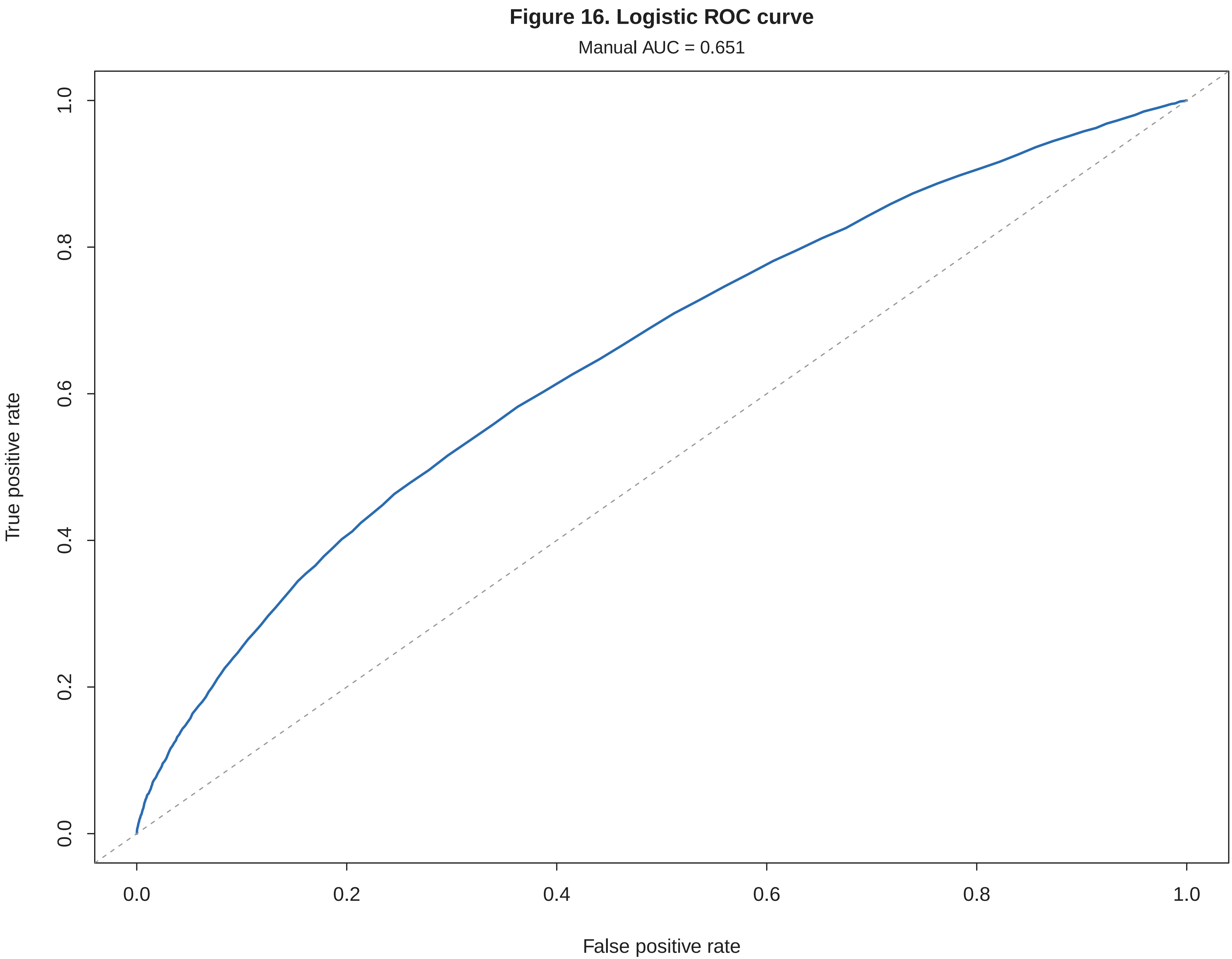
Table 14. Predicted probability example

time_in_hospital	num_lab_procedures	num_procedures	num_medications
1	41	0	1

number_outpatient	number_emergency	number_inpatient	number_diagnoses	race
0	0	0	1	Caucasian

gender	age	Predicted_Probability
Female	[0−10)	0.1476

The default 0.5 threshold is transparent but may under−detect readmissions if the positive class is less common. Clinical screening would normally compare thresholds using costs of false positives and false negatives.



Caption: ROC curve summarizes ranking performance across thresholds.

Final Interpretation and Conclusion

USA Housing findings

The multiple linear regression explains 91.8% of variation in Price (adjusted R−squared 0.918). Holding the other model variables constant, positive coefficients indicate higher expected price. The largest coefficient by scale is Avg_Area_House_Age, but variable units differ, so practical importance should be judged with context. Statistically significant predictors at $p < 0.05$ are: Area_Population, Avg_Area_House_Age, Avg_Area_Income, Avg_Area_Number_of_Rooms.

Diabetes readmission findings

The logistic model estimates the probability of readmission. Odds ratios above 1 indicate higher odds of readmission, while odds ratios below 1 indicate lower odds. At threshold 0.5, test accuracy is 61.8%, sensitivity is 38.9%, specificity is 81.4%, and manual AUC is 0.651. Statistically significant predictors at $p < 0.05$ include: age[10−20), age[20−30), age[30−40), age[40−50), age[50−60), age[60−70), age[70−80), age[80−90), genderMale, num_lab_procedures, num_medications, num_procedures, number_diagnoses, number_emergency, number_inpatient, number_outpatient, raceCaucasian, raceOther, raceUnknown, time_in_hospital.

Limitations

Both analyses are observational and should not be interpreted causally. The housing dataset uses area−level variables, not full property−level appraisal data. The diabetes model uses coded administrative fields selectively; high−cardinality specialty and *_id variables can create unstable or hard−to−interpret coefficients. Classification at a 0.5 threshold may not be optimal for clinical readmission screening.

Recommended next analytical steps

Next steps include validating models on external data, testing nonlinear effects and interactions, evaluating alternative probability thresholds for readmission, and comparing base regression with regularized or tree−based models after documenting package dependencies.